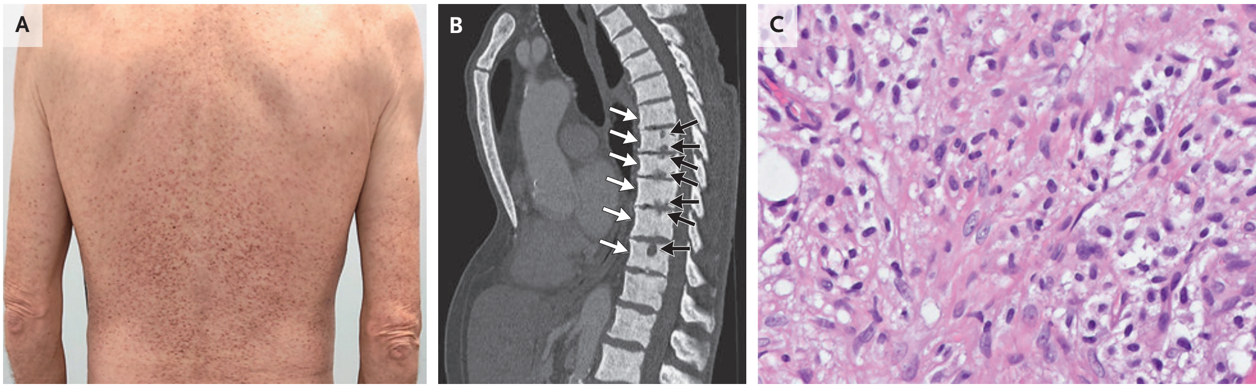
IMAGES IN CLINICAL MEDICINE

Systemic Mastocytosis



A 65-YEAR-OLD MAN PRESENTED TO THE INTERNAL MEDICINE CLINIC WITH a 4-month history of a rash and a 2-month history of diarrhea and unintentional weight loss. Physical examination was notable for a maculopapular eruption on the trunk (Panel A), arms, and legs. Laboratory testing showed normocytic anemia and a serum tryptase level of more than 200 μg per liter (reference value, <11). Computed tomography of the abdomen and pelvis showed osteosclerosis of the thoracic vertebral bodies (Panel B, white arrows; sagittal view) and erosions of the end plates (Panel B, black arrows). No hepatosplenomegaly was seen. Skin biopsy revealed mast-cell infiltration. Bone marrow biopsy revealed cohesive groups of round and spindle-shaped cells with dense chromatin and eosinophilic cytoplasm and marked collagen fibrosis (Panel C, hematoxylin and eosin stain). Immunohistochemical staining was positive for CD25 and CD117 (c-Kit), which regulates mast-cell development and is mutated in a variety of cancers. Molecular testing identified the gain-of-function D816V mutation in *KIT*. A diagnosis of systemic mastocytosis — a condition characterized by mast-cell infiltration of organs, such as the bone marrow, liver, gastrointestinal tract, and cortical bone — was made. Treatment with antihistamines and the c-Kit inhibitor midostaurin was initiated. At 1 year of follow-up, the patient's symptoms had abated.

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